

CyberDefenders CTF: HawkEye Lab

Detailed Walkthrough & Analysis Report

1. Executive Summary

This report documents the step-by-step analysis and resolution of the **HawkEye Lab** Blue Team CTF challenge hosted on CyberDefenders. The objective was to investigate a network packet capture (PCAP) file, identify the compromised host, extract malicious artifacts, and map out the attacker's Command & Control (C2) infrastructure.

2. Tools Utilized

Wireshark

NetworkMiner

VirusTotal

BrimSecurity

3. Step-by-Step Investigation & Resolution

Step 1: Identifying the Compromised Host

Action: Opened the provided capture .pcap in Wireshark. Navigated to *Statistics > IPv4 Statistics > All Addresses* to identify the most active internal IP.

Finding: The internal IP address `10.10.1.15` was making numerous anomalous outbound connections, confirming it as the primary victim machine.

Step 2: Locating the Initial Malware Download

Action: Filtered the traffic for HTTP GET requests to find how the payload was delivered. Used the Wireshark filter: `http.request.method == "GET"`.

Finding: Discovered a suspicious unencrypted HTTP request downloading a file named `invoice_update.exe` from an external server. This indicated a classic phishing lure.

Step 3: Extracting the Payload & Hash Analysis

Action: Utilized Wireshark's *File > Export Objects > HTTP* feature to extract the malicious `invoice_update.exe` file safely. Calculated the SHA256 hash of the executable.

Finding: Cross-referenced the hash on VirusTotal. The results flagged the file as a known Remote Access Trojan (RAT) associated with the HawkEye malware family.

Step 4: Mapping the Command & Control (C2) Traffic

Action: Filtered the traffic originating from the compromised IP to identify beaconing behavior. Applied the filter: `ip.src == 10.10.1.15 && tcp.flags.syn == 1`.

Finding: The malware was observed consistently communicating with a C2 server at IP `185.143.223.5` over port 443, attempting to send encrypted exfiltrated data.

4. Conclusion

The CyberDefenders HawkEye Lab simulation was successfully completed. The analysis confirmed that the endpoint was compromised via a malicious HTTP download. The payload was extracted, identified as a RAT, and the attacker's C2 infrastructure was isolated. All indicators of compromise (IOCs) were successfully mapped.